Practical 1:

* Problem statement:

-> WAP to create an array and demonstrate insertion, deletion, traversal, and search operations.

* Algorithm:

1. Start

2. Create an Array

- Read the number of elements n.
- Input n elements into the array arr.

3. Traversal operation

- Display all elements of the array from the first element to the last element.

4. Insertion operation

- Read the position pos where the element is to be inserted.
- Read the value item to be inserted.
- Shift all elements from position pos onwards one position to the right.

- Insert item at Position Pos.
- Display the updated array.

5. Deletion operation

- Read the position Pos of the element to be deleted.
- Shift all elements after Pos one Position to the left.
- Delete the element at Position Pos.
- Display the updated array.

6. Search operation

- Read the element key to be searched.
- Traverse the array from the first element to the last element.
- If key is found:
  • Display the Position of element
  • Stop searching
- Otherwise, display "Element not found."

7. Stop.

* code :

```
#include <iostream>
using namespace std;

int main() {
    int arr[], n;

 // Array creation

    cout << "Enter number of elements:
     ";
      cin >> n;

     cout << "Enter elements:\n";
      for (int i = 0; i < n ; i++) {
          cin >> arr[i];
      }

// Traversal

   cout << "Array elements are: " << endl;

    for (int i = 0; i < n; i++) {
        cout << arr[i] << " ";
   }
```

// Insertion

```
int Pos, ~~item~~ value;

cout<<"Enter index to insert :"<<endl;
cin >> Pos;

cout << "Enter value to insert:"
        <<endl;
cin >> ~~item~~; value;

for (int i=n ; i> Pos ; i--) {
    arr[i] = arr[i-1];
}
arr[Pos] = item;
n++;

cout << "Array after insertion:";

for (int i=0 ; i<n ; i++) {
    cout << arr[i] << " ";
}
```

// Deletion

```
cout<<"\n\n Enter index to delete:
        ";
cin >> Pos;
```

```
for (int i = pos; i<n-1; i++) {
        arr[i] = arr[i+1];
  }
        n--;
    cout << "Array after deletion:";

    for (int i = 0; i<n; i++) {
          cout << arr[i] << " ";
    }

// search
  int key, ~~found = 0~~, found = 0;

  for (int i = 0; i<n; i++) {
        if (arr[i] == key) {
     cout << "Element found at
            index:" << i << endl;
          found = 1; break;
    }
}
     if (!found) {
         cout << "Element not found"
                    << endl;
     }
     return 0;
}
```

⟹ Output:

Enter number of elements : 4
Enter elements:
1 2 4 5

Array elements are : 1 2 4 5

Enter index to insert : 2
Enter value to insert : 3
~~After~~ Array after insertion:
1 2 3 4 5

Enter index to delete : 4
Array after deletion : 1 2 3 4

Enter element to search : 4
Element found at index : 3